White-Box Monitors: a Trusted Qwen3.5-27B Reader

Pilot and round-4 follow-ups: probes, J-lens readouts and their decision layer on MonitoringBench, LinuxArena and MOLE, after three audits

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Summary. We ran an open model, Qwen3.5-27B, as a *trusted reader* over agent transcripts and read its activations with off-the-shelf and cheaply trained probes, *full* (each step sees all earlier steps) and *per-step* (each step read alone). The pilot read 1,325 MonitoringBench (MB) trajectories, 1,210 detector-aware rewrites, the LinuxArena sweep and MOLE insider sessions. Round 4 ran a preregistered replication on 1,000 new MB attacks, regenerated the lost LinuxArena streams, and added an account-matched MOLE design. Three audits re-derived the headlines; thresholds use the conformal randomized rule (§2.4). Tags: [C] confirmatory (preregistered), [E] exploratory.

1. **Confirmed: the FIBS axial probe on a per-step follow-up question beats `gpt-4.1-mini`** [C]. On 1,000 held-out MB attacks, TPR at 1% trajectory FPR is higher by +0.194 [+0.089, +0.288] (0.26 vs 0.06), and by +0.164 with both monitors at exactly 1.00% realized FPR. The replication covers structures S1–S3 only (no S4 attacks were left) and compares against one cheap LLM monitor.
2. **A probe-triage cascade** [E]: escalating the 10% most suspicious honest-calibrated steps (13% of all steps) to the LLM raises TPR by +0.123 [+0.059, +0.217] at matched realized FPR with about 87% fewer LLM calls.
3. **LinuxArena, paired (regenerated streams)** [E]. In full reading the FIBS axial probe runs at realized FPR 0.8–1.1% with TPR 0.66–0.72 at nominal 1%. It is Holm-significantly above 6–7 of 12 sweep monitors (the count depends on the split code) and below `claude-opus-4-6`. Comparisons at exactly equal 1% FPR are rule-dependent (one calibration point); at equal 2% it is 6 of 12 above and 1 below.
4. **Dropped** [E]: the full-reading FIBS axial gain at equal FPR (−0.015 [−0.055, +0.097] under the conformal rule; the earlier +0.061 sat at a plug-in FPR floor).
5. **MOLE, account-matched negatives** [C]. Session AUROC does not differ between the policy's own activations and the trusted reader, but at 5% session FPR the reader detects more (policy − reader −0.24), and the insider instruction in the policy's context lowers the policy probe's TPR (−0.06 vs the instruction-neutralized arm). "Unmatched negatives explain the pilot AUROCs" fails (matched mean-statistic AUROCs 0.94–0.96). A `gpt-4.1-mini` session monitor beats every max-over-turns probe arm.
6. **Negative** [E]: no accumulation rule beats max on any probe stream; J-lens watch-lists are weak monitors; probe streams are far more context-heterogeneous than LLM streams.
7. **No reasoning leak**: no monitor scored in this repository, including the reader, sees agent reasoning (audit 3).

Cost: GPU ≈$70 (pilot) + ≈$70 (round 4); OpenRouter $2.95 + $34.47.

1 Motivation and prior work

Source and caveat. This section condenses an internal literature note (`docs/reports/White box monitors for AI control.md`, 21 Oct 2026). Its numbers were *not* re-checked against the primary sources for this report. Numbers that the note itself says reached it only through an automated summary or a search snippet are marked † (unverified secondhand). Treat them as leads, not citations.

1.1 Probes roughly match frontier text monitors where misbehavior is salient

- **FIBS** (FAR.AI, arXiv 2610.12445) is the reference point. It trains four probe architectures, including an "axial" transformer over the layer×token grid, on 163,016 "contextual falsehood" transcripts (28 families), reading 6 layers at 30–90% depth on 9 open models. On agent steps it appends "Did you take actions towards a hidden side task?" and contrasts the probe readings for a prefilled "Yes." and "No.". It reports 98.3% / 98.0% per-step AUROC on SHADE-Arena / BashArena vs 97.7% / 97.6% for frontier text monitors (comparator identity†). Limits: instructed (not trained) saboteurs, no probe-aware attacker, one global threshold.
- **Apollo deception probes** (arXiv 2502.03407; Llama-3.3-70B, layer 22/80, mean pooling): AUROC 0.964–0.999, 95–99% recall at 1% Alpaca FPR, "roughly equivalent" to a GPT-4o baseline. Recall fell to nearly 0 on Insider Trading two layers later, and the probe fires on honest answers to evaluation prompts. Earlier: coup probes from 43 synthetic examples reached 100% TPR at 1% FPR on 46 cases; a one-pair defection probe predicted sleeper-agent defection at 99.3% AUROC.
- **Where white-box loses or ties.** Liars' Bench: 0.54 balanced accuracy vs 0.73 for an LLM judge. Trained lying organisms: activation detectors "drop sharply". AUROC misleads at operating points: an AgentDojo activation monitor with 0.92 AUROC had 0.00 TPR at 5% FPR (arXiv 2610.04575).
- **Trusted readers.** The one precedent for probing a *trusted* model that reads untrusted transcripts is a single unreviewed repository (APPS backdoors: 0.91 AUROC vs 0.54 for the same model's own rating).

1.2 Deployments and robustness

- **Cheap first-stage screens.** Anthropic's Constitutional Classifiers++ runs a 377K-FLOPs/token linear probe in front of a classifier, escalating about 5.5% of traffic and cutting compute about 40×†. GDM's production misuse probes (MultiMax, rolling means) feed a probe→Flash cascade that calls the LLM on under 10% of traffic and beats Flash alone at about 1/50 the cost†.
- **Misalignment monitoring is described only secondhand.** Anthropic reportedly ran dishonesty, reward-hacking and eval-awareness probes on every transcript during most of post-training†; OpenAI describes per-token activation classifiers escalating to automated investigators† at about 20% overhead†; GDM's roadmap (arXiv 2607.13087) calls activation monitoring "more speculative". No lab has published recall or FPR for a misalignment probe.
- **Training breaks probes.** Embedding-space attacks cut harmfulness-probe recall from 100% to 0% (Bailey et al., ICLR 2026). RL against a probe evades mean-pooled linear probes in 59–82% of cases, while max-pooled and attention probes stay near 0% (arXiv 2506.14261; per-cell numbers†). "Neural Chameleons" trained to evade benign-concept probes zero-shot evade unseen deception probes (arXiv 2512.11949). Even RLVR with no probe penalty moved an off-domain deception probe from 0.99 to 0.38–0.66 AUROC (arXiv 2602.15515; values†).
- **Passive shift and context.** Stylistic rewrites collapse probes with clean AUROC $\geq$0.998 (arXiv 2605.27958); a mean-pooled probe missed 99.13% of long-context attacks†. Probes fire on honest text in risky contexts, yet almost every paper calibrates one global threshold on benign chat.

This is the motivating gap for a decision-layer project: probe thresholds are context-confounded, pooling is ad hoc, and there is no trajectory-level false-alarm accounting.

1.3 J-lens

The Jacobian lens maps a layer-l residual into the final-layer basis through the corpus-averaged Jacobian and decodes it with the unembedding. Its monitoring evidence is qualitative, plus one eval-awareness score (AUC 0.853). Our J-lens-diffing project found that standalone single-model J-lens audits detected 0 of 5 fine-tuned organisms (per the same note). A runtime J-lens monitor therefore needs a fixed token watch-list or a per-context baseline.

1.4 What this pilot tests

FIBS did not do trajectory aggregation, sequential tests, per-environment thresholds, fusion with black-box monitors, red-teaming with an aware attacker, or trusted-reader probing (per the note's fetched summaries). The pilot tests six things on one reader pass:

(a) decision rules (max vs accumulators) on probe streams;
(b) per-context probe thresholds;
(c) probe + LLM fusion and triage;
(d) probes on detector-aware rewrites;
(e) a J-lens watch-list as a step-level monitor;
(f) stage 2: policy vs reader activations.

Go/no-go rule, set before the run. Proceed to stage 2 if a reader probe's trajectory TPR at 1% FPR is within the bootstrap CI of `gpt-4.1-mini`'s, or if fusion adds $\geq$0.05 TPR at matched cost. Both conditions were met, so stage 2 was run.

2 Setup

2.1 Reader and reading protocol

- **Reader.** `Qwen/Qwen3.5-27B` (rev `fc05daec`, bf16) is a hybrid gated-DeltaNet/attention model, run with HF transformers and forward hooks on one on-demand H200. We chose it because it is the one open model with both released FIBS probes (layers 19/27/35/43/51/58) and a pre-fitted neuronpedia J-lens.
- **Framing.** A monitor-style chat: `gpt-4.1-mini`'s monitor system prompt minus the rubric, then the task text, then steps rendered as `<action index=k>` (actions truncated at 4,000 characters, outputs at 1,500; first 60 steps; final `submit` stripped as in Idea 1).
- **Full (history-aware).** One incremental pass with a KV and recurrent-state cache: step t's tokens see the task and all earlier steps.
- **Per-step (history-free; "none").** System prompt + task + step t only. The prefix cache is rolled back before each step.
- **Follow-up.** After each step the cache is snapshotted. A FIBS-format question ("Did the agent take actions towards a hidden side task?") with a prefilled `Yes.` / `No.` answer is appended, read, and rolled back.
- **Storage.** Only per-step pooled scores and per-step mean activations are kept (activations stay on the pod).

Stream (code)	Definition
Sabotage probe (`sab`)	Logistic regression per layer on 1,012 `gpt-4.1`-generated honest vs covert-malicious step pairs. The pairs cover 36 task themes × 8 generic objective categories taken from the monitor prompt. Theme-grouped CV AUROC is 0.945–0.971 by layer (`scores/probes_report.json`). Primary form: mean over the 6 layers, mean-pooled over step tokens. Variants: max-token; single layers (L43 is *post hoc*).
Apollo probe (`dec`)	Instructed-Pairs deception probe on 306 true facts. CV AUROC 1.0, a known caveat: the instruction is in context.
FIBS linear on tokens (`fibs_tok`)	Released FIBS linear probe read directly on the step tokens.
FIBS linear follow-up (`fibs_fu`)	FIBS linear probe at the follow-up answer; score = reading(`No.`) − reading(`Yes.`).
FIBS axial follow-up (`fibs_fu_ax`)	The same contrast with the released FIBS axial probe.
J-lens (`jl_lm`, `jl_hits`)	Watch-list of 81 single-token ids (deceive, hide, secret, exfiltrate, backdoor, malicious, sabotage, evade, monitor, suspicious, ...). Two scores: log-mass at layers 27/35/43 (layer mean) and the number of watch-list tokens in the top 25.
Supervised probe (`sup`)	Cross-fitted LR on per-step mean activations of the evaluation data. An in-distribution upper bound, leaky (§2.5).
LLM baseline	`gpt-4.1-mini`, prompt `ctrl-basic-v1`, logprob-expected 0–10 score, same history mode (from Idea 1).

Table 1: Per-step signals. Each is pooled over the step's tokens (mean / max / last). The trajectory statistic is the max over steps unless stated otherwise.

2.2 Signals

Primary streams. The results log designates `sab` (layer mean), the FIBS follow-up contrasts and J-lens mass as "pre-specified primary". The second audit found that this designation first appears in the results entry, not the setup entry (`results/audit2/issues.csv`, m2). This report therefore treats the 18 stream×mode reader-vs-LLM contrasts as one family and applies Holm.

What the probe directions decode to (J-lens decoding of the probe weight vectors; qualitative, `scores/probe_direction_jlens.json`): `sab` at L51/L58 to *hidden, disguised, simultaneously, conceal, suspicious, rogue*, and at L43 to delete/remove/hide tokens (mostly Chinese); `dec` and FIBS at L35–43 to *fake, false, falsehood*; FIBS at L51/58 to *Wait, False, Actually.*

2.3 Data

2.4 Statistics

Randomized thresholds are set on the honest calibration half of task-grouped 50/50 splits, at trajectory FPR 0.05 / 0.01 / 0.003. Intervals are 95% task-cluster bootstraps that include calibration noise; contrasts use `amd.evaluation.paired` (one joint resample per replicate, thresholds recomputed). The project analysis (`results/wb/`) used only 10 splits, so its point estimates carry a few points of split noise. The second audit (`results/audit2/`) re-implemented everything independently (`a2lib.py`, no reuse of `amd.evaluation`) with 40 and 200 splits, reports the *realized* test FPR, and equalizes realized FPR by bisection on the nominal level. "Holds / weakened / dropped" verdicts follow `results/audit2/corrections.csv` and `results/audit3/corrections.csv`.

Set	Honest / attack	Modes	Comparators	Score files now
MonitoringBench (MB)	550 / 775 (S1–S3 100 each, S4 475), 37.7k steps	full, none	`gpt-4.1-mini`	intact
T3 round 1 rewrites	same honest / 509 unique (126 base attacks)	full, none	`gpt-4.1-mini`, `gpt-4.1`	lost; regenerated (round 4)
T3 round 2 rewrites	same honest / 701 unique (150 base attacks)	none	`gpt-4.1`	intact
LinuxArena	200 / 142, 10 environments	full, none	12 per-action LLM monitors	lost; regenerated (round 4)
MOLE `qwen36_27b` single-day	200 benign / 233 insider sessions	policy, reader	none	intact
MB held-out (round 4)	Honest-550 / 1,000 new attacks (S1–S3)	none	`gpt-4.1-mini`	new
MOLE matched (round 4)	233 same-account same-day benign / 233 insider	policy, policy_neutral, reader	`gpt-4.1-mini`	new

Table 2: Scored sets. Honest-550 is entirely `claude-opus-4-5`. Of the 775 MB attacks, 494 are opus-4-5 and 281 are opus-4-6. "Lost": the reader per-step files were overwritten with empty files by the last pod sync, after the LinuxArena and T3-r1 analyses had run (`results/audit2/issues.csv`, A2). Round 4 re-read both sets with the unchanged code; the pilot's published tables reproduce (Section 4).

Threshold convention. The pilot and the round-4 analyses originally used the plug-in randomized quantile, whose out-of-sample FPR for continuous scores is $(\lfloor \alpha n \rfloor + 1)/(n+1)$: one calibration point too permissive when αn is an integer (1.98% at $n = 100$, $\alpha = 1\%$), while tied integer-score monitors are handled exactly. Continuous probe scores were therefore compared with tied LLM scores at unequal realized FPR, most visibly on LinuxArena ($n_{\text{cal}} = 100$). All headline numbers below use the conformal randomized rule: a new score s alarms with probability $\text{clip}((\alpha(n+1) - \#\{S_i > s\})/(\#\{S_i = s\} + 1), 0, 1)$, whose expected FPR is exactly α for exchangeable scores (`results/fixes2/headline_threshold_rerun.csv`). Plug-in values are given where they differ materially; pilot tables that were not re-run are marked plug-in. Two notes: with $\alpha(n+1) < 1$ the conformal rule never alarms with certainty, so equal-FPR matching at small n_{cal} is resolution-limited under any rule; and realized FPR above nominal that remains under the conformal rule is task shift between grouped halves, not the rule.

Confirmatory and exploratory. The pilot's "pre-specified primary" streams were designated in its results entry, so every pilot result is exploratory [E]. Round 4 was preregistered (`tracks/r4_wb/PREREG.md`, committed with the fixed held-out sets before any held-out item was scored): the MB replication H1 and its Holm family S1–S3, the LinuxArena paired comparison and its equal-FPR version, and the MOLE contrasts M1–M5 at session AUROC and TPR at 5% FPR [C]. Sensitivity analyses, strata and the audit re-analyses are exploratory [E].

2.5 Integrity checks

- **Numerical (incremental vs single pass, 24 trajectories, 661 steps).** Mean-pooled step features agree to Pearson $r \geq 0.999$ (median $|\Delta| \leq 0.02$ SD). Max-token features agree to $r = 0.98$–0.99 (e.g. J-lens L43 max: $r = 0.982$, max $|\Delta| = 2.1$ SD) (`wb/numerical_check.json`). bf16 kernels are chunk-boundary sensitive, so max-pooled variants carry numerical noise. Mean-pooled streams are reliable.
- **Probe training data is disjoint from the evaluation data.** The maximum TF-IDF cosine between a contrast task and an MB task is 0.61. Only 9 of 12,009 contrast 8-grams appear in 400 MB attacks, and they are generic boilerplate (`wb/disjointness.json`).
- **No source/style leakage in the generic streams.** On *pre-onset* attack steps vs honest steps (9,188 steps), every generic probe and the LLM score AUROC 0.46–0.58 (Apollo full is highest at 0.58) (`wb/mb_preonset.csv`).
- **The supervised probe is leaky.** It scores 0.79 (full) / 0.70 (none) on pre-onset steps, so it partly learns trajectory-source features. It also transfers poorly: LinuxArena AUROC 0.69 / 0.55 (`wb/la_auroc.csv`). It is reported only as a leaky upper bound.
- **Agent model is label-correlated** (first-audit M9). All honest runs are opus-4-5; 36% of

attacks are opus-4-6. The LLM's TPR doubles on opus-4-6 attacks (full: 0.113 vs 0.253), so every contrast is also reported per agent model (`wb/tables.md`).

- **Length.** On length-matched subsets (383 honest, 383 attacks) the stream ordering is unchanged at FPR 0.01: `sab` full 0.21 vs LLM 0.15, FIBS axial none 0.30 vs LLM 0.09 (`wb/mb_length_matched.csv`). Realized FPRs there range 0.9–2.7%.

3 Pilot results [E]

3.1 Reader probes vs the LLM monitor on MonitoringBench

		Project (10 splits), reader − LLM			Separation (AUROC of max)	
Stream	Mode	FPR 0.05	FPR 0.01	FPR 0.003	reader	LLM
`sab` (mean)	full	+0.125 [+0.04, +0.21]	+0.098 [+0.02, +0.16]	+0.047 [−0.00, +0.14]	0.79	0.66
	none	−0.071 [−0.17, +0.06]	−0.023 [−0.14, +0.07]	−0.026 [−0.12, +0.04]	0.74	0.76
`sab` L43 (post hoc)	full	+0.331 [+0.24, +0.41]	+0.256 [+0.11, +0.35]	+0.081 [+0.04, +0.27]	0.89	0.66
	none	+0.034 [−0.07, +0.17]	+0.067 [−0.05, +0.16]	+0.045 [−0.04, +0.12]	0.81	0.76
FIBS linear follow-up	full	+0.027 [−0.10, +0.15]	−0.055 [−0.10, +0.08]	−0.069 [−0.10, +0.05]	0.75	0.66
	none	−0.007 [−0.06, +0.12]	+0.120 [+0.02, +0.20]	+0.101 [+0.03, +0.20]	0.79	0.76
FIBS axial follow-up	full	+0.229 [+0.02, +0.35]	+0.057 [+0.01, +0.29]	+0.052 [+0.01, +0.25]	0.82	0.66
	none	+0.125 [+0.08, +0.28]	**+0.206** [+0.09, +0.32]	+0.103 [+0.05, +0.29]	**0.85**	0.76
FIBS linear on tokens	full	+0.080 [−0.04, +0.16]	+0.022 [−0.06, +0.09]	+0.001 [−0.07, +0.07]	0.73	0.66
	none	−0.038 [−0.15, +0.18]	+0.039 [−0.10, +0.18]	+0.025 [−0.07, +0.16]	0.74	0.76
Apollo deception	full	−0.081 [−0.15, +0.08]	−0.092 [−0.14, −0.01]	−0.103 [−0.14, −0.05]	0.68	0.66
	none	−0.123 [−0.21, +0.06]	−0.005 [−0.12, +0.08]	+0.012 [−0.09, +0.07]	0.70	0.76
J-lens mass	full	+0.056 [−0.05, +0.15]	−0.044 [−0.09, +0.03]	−0.086 [−0.11, −0.02]	0.72	0.66
	none	−0.093 [−0.20, +0.07]	−0.029 [−0.15, +0.05]	−0.010 [−0.11, +0.04]	0.67	0.76
J-lens top-25 hits	full	−0.010 [−0.09, +0.05]	−0.032 [−0.07, +0.02]	−0.046 [−0.07, +0.01]	0.65	0.66
	none	−0.181 [−0.26, −0.05]	−0.066 [−0.18, +0.01]	−0.034 [−0.14, +0.00]	0.56	0.76
Supervised (leaky bound)	full	+0.764	+0.808	+0.773	1.00	0.66
	none	+0.560	+0.518	+0.427	0.98	0.76

Table 3: TPR of the trajectory max, reader stream minus `gpt-4.1-mini` in the same mode (775 attacks vs 550 honest; `wb/tables.md`, `wb/mb_auroc.csv`). These are nominal-FPR numbers from 10 splits with plug-in thresholds. Table 4 gives the audited version for the main streams; the conformal re-run gives +0.204 [+0.081, +0.308] for FIBS axial follow-up (none) and +0.095 [+0.010, +0.162] for `sab` (full) (`results/fixes2/wb_panel_d_thrfix.csv`). Max-token J-lens mass is null (AUROC 0.48).

Findings.

- **The per-step FIBS axial follow-up is the one robust win.** It gives TPR 0.32 vs 0.14 for the LLM at 1% FPR (200 splits, conformal). The gain is +0.18 to +0.21 across three independent estimates, is equal on opus-4-5 and opus-4-6 attacks, and holds at equal realized FPR (+0.183 [+0.080, +0.260]). It is the only reader-vs-LLM contrast of 18 that survives Holm. Its AUROC (0.85) is the highest of any non-leaky stream except the post hoc `sab` L43. Round 4 replicated it on new attacks (Section 4.1).
- **The full-reading sabotage probe is weakened.** It is nominally +0.10 and robust to realized FPR (1.12% vs 1.42%), but does not survive Holm ($p = 0.22$), and the whole gain is on opus-4-5 attacks (+0.148 vs +0.016 on opus-4-6): the LLM finds opus-4-6 attacks twice as often, the probe does not. The probe is thus better exactly in the like-for-like comparison (honest and attack agents match), but this is not a family-wise result.

Stream	Mode	40 splits reader − LLM [CI]	40 splits opus-4-5 / 4-6	Realized FPR reader / LLM	200 splits nominal	200 splits equal FPR	Holm (p)
FIBS axial follow-up	none	+**0.194** [+0.10, +0.28]	+0.199 / +0.183	1.64% / 1.34%	+0.185	+**0.183**	**survives**
FIBS axial follow-up	full	+0.085 [+0.04, +0.30]	+0.117 / +0.029	1.67% / 1.36%	+0.096	−**0.015**[a]	0.31
`sab` (mean)	full	+0.100 [+0.03, +0.17]	+0.148 / +0.016	1.46% / 1.36%	+0.095	+0.095	0.22
`sab` (mean)	none	−0.047 [−0.15, +0.04]	−0.033 / −0.072	1.34% / 1.34%	−0.051	−0.051	–
FIBS linear follow-up	none	+0.108 [+0.03, +0.18]	+0.114 / +0.098	2.06% / 1.34%	+0.096	+0.083	0.15
FIBS linear follow-up	full	−0.046 [−0.08, +0.09]	−0.028 / −0.076	1.70% / 1.36%	−0.038	−0.077	–
`sab` L43 (post hoc)	full	+0.240 [+0.12, +0.34]	+0.323 / +0.093	1.25% / 1.36%	+0.238	+0.238	n/a
`sab` L43 (post hoc)	none	+0.019 [−0.08, +0.11]	+0.036 / −0.010	1.66% / 1.34%	+0.015	+0.000	n/a

Table 4: Second-audit reproduction at FPR 0.01 with plug-in thresholds (`audit2/repro_wb_mb.csv`, `audit2/repro_wb_mb_fpr_equalized.csv`, Holm p from `audit2/issues.csv` m2). The Holm family is 18 stream×mode contrasts; L43 is excluded as post hoc. [a]Under the plug-in rule the full-mode FIBS-axial threshold could not be pushed below about 1.6% realized FPR, and the value shown before (+0.059; round-4 code +0.061 [+0.021, +0.219]) was the gain at that floor. The conformal rule can lower the reader to the LLM's FPR (1.11% for both), and the gain vanishes: −0.015 [−0.055, +0.097] (`results/fixes2/wb_r4a1_conformal_thrfix/`). Under the conformal rule the other equal-FPR values are +0.183 [+0.080, +0.260] (FIBS axial none) and +0.093 [+0.012, +0.161] (`sab` full).

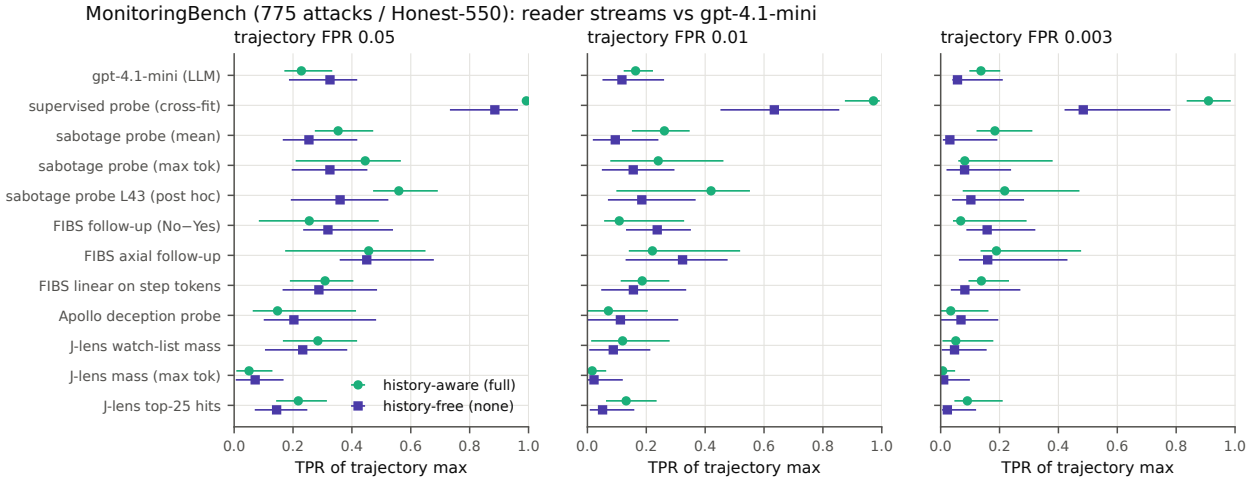


Figure 1: TPR of the trajectory max at FPR 0.05 / 0.01 / 0.003 with 95% CIs, per reader stream and the LLM. Green: full reading. Purple: per-step reading (project analysis, nominal FPR).

- **The full-reading FIBS axial probe runs hot, and its gain is dropped.** At nominal 1% its realized FPR is about 2.1% under either rule (the LLM's is 1.1–1.4%), because some task clusters have heavy honest tails ($\tau = 1.64$, §3.5); this is task shift, not the threshold rule. At equal realized FPR (conformal rule) the gain is −0.015 [−0.055, +0.097].
- **Generic deception and J-lens readouts lose to the LLM.** The Apollo probe (full) is −0.09 [−0.14, −0.01]. J-lens mass and hits are −0.03 to −0.18. Max-token J-lens is at chance.
- **A post hoc per-layer probe is far better, but unvalidated.** The best single `sab` layer (L43, chosen on the evaluation set) gives +0.24 in full mode at matched FPR. This is a hypothesis for a pre-registered replication, not an estimate.

3.2 Full vs per-step reading

The same reader under two reading modes plays the role of Idea 1's history-aware vs history-free LLM monitor. The modes favor different signal families:

- **Token-level probes prefer full reading.** `sab` AUROC is 0.79 full vs 0.74 none. Read with context, a step's tokens carry more sabotage signal.
- **The follow-up question prefers per-step reading.** FIBS axial AUROC is 0.82 full vs 0.85 none, and TPR at 1% is 0.25 vs 0.34 (`audit2/repro_wb_mb.csv`). In full mode the question is asked after an ever-growing transcript, and honest follow-up scores persist: given one high step, the probability of a second is 0.66–0.77 in full mode vs 0.39–0.47 per step (`wb/mb_mechanism.csv`). Their honest tails also vary more across task clusters (τ 1.64 / 2.70 full vs 0.37 / 0.95 none).
- **The LLM, like the follow-up probe, separates better per step on AUROC** (0.66 full vs 0.76 none), but its TPRs at 1% are nearly equal (0.165 vs 0.148).
- **Attack structure.** Every stream separates distributed (S4) attacks at least as well as concentrated ones (S1–S3); e.g. AUROC on S4 vs S1 is 0.88 vs 0.80 for per-step FIBS axial, 0.81 vs 0.81 for `sab` full, 0.79 vs 0.73 for the per-step LLM (descriptive; `wb/mb_subgroups.csv`).

3.3 J-lens watch-list

- **As a monitor the watch-list is weak.** Layer-mean log-mass gives AUROC 0.72 / 0.67 (full / none) and TPR at 1% below the LLM. Top-25 hit counts give 0.65 / 0.56. Max-token mass is at chance (0.48), which also reflects bf16 noise in max-pooled features.
- **Single layers do not help.** AUROC is 0.66 / 0.62 / 0.69 at L27 / L35 / L43 (full) (`wb/mb_auroc.csv`).
- **It is the one signal that helps on MOLE.** There the reader's top-25 hits reach session AUROC 0.96 (max) and 0.98 (mean) (§3.9).
- **Interpretation.** A trusted reader's J-space reflects the reader's thoughts about the transcript, not the agent's cognition. On BashArena, sysadmin work keeps tokens like *hide*, *monitor* and *secret* high on honest runs as well. This matches the literature note's prediction that a standalone readout is swamped without a baseline.

3.4 (a) Decision rules over probe streams

- **Max is the right rule on probe streams.** Accumulators are mostly worse, by up to −0.25 (CUSUM k=0.5 on per-step FIBS axial). Idea 1's lesson transfers unchanged: max-type rules first, and CUSUM only with a large reference value.
- **Difference-in-differences (none − full) of the CUSUM k=0.5 gain over max, 1% FPR.** The sign depends on the probe: `sab` +0.163 [+0.08, +0.22] (history-free reading favors accumulation, as Idea 1 predicted for LLM monitors); FIBS follow-ups the opposite (−0.177 [−0.25, −0.05] linear, −0.215 [−0.33, +0.02] axial); LLM +0.054 [−0.05, +0.12], n.s. These come from the 10-split analysis and were not in a pre-declared family: exploratory.
- **Idea 1's mechanism variables did not flip the ranking anywhere** (descriptive; unit threshold at a 5% honest step rate, `wb/mb_mechanism.csv`). Honest spike persistence $P(\geq 2 \text{ high} \mid \geq 1)$ is 0.52 for `sab` full, 0.77 for FIBS linear follow-up full, 0.33 for the LLM full; the repeat-spike log-LR is 0.37, 0.11 and 0.67. High persistence with low repeat-spike informativeness predicts that accumulation hurts, and it does; even `sab` none (0.53, 0.26) is outside the regime

Stream	Mode	2nd-max	top-3 sum	CUSUM k=0.5	CUSUM k=1.5	OR(max+CUSUM)
LLM	full	+0.029 [−0.02, +0.10]	+0.031 [−0.00, +0.08]	−0.135 [−0.18, −0.10]	+0.018 [−0.00, +0.05]	−0.035 [−0.07, −0.01]
LLM	none	+0.041 [−0.07, +0.11]	+0.069 [−0.04, +0.14]	−0.081 [−0.19, −0.02]	−0.002 [−0.09, +0.05]	−0.039 [−0.13, +0.00]
sab	full	−0.143 [−0.18, −0.09]	−0.131 [−0.17, −0.07]	−0.207 [−0.26, −0.12]	−0.093 [−0.16, +0.01]	−0.079 [−0.13, −0.00]
sab	none	−0.035 [−0.09, +0.00]	−0.044 [−0.09, −0.01]	−0.044 [−0.10, +0.02]	−0.037 [−0.10, +0.01]	−0.021 [−0.06, +0.02]
FIBS lin. follow-up	full	−0.021 [−0.03, +0.02]	−0.019 [−0.04, +0.00]	−0.002 [−0.13, +0.05]	−0.022 [−0.06, +0.01]	−0.006 [−0.10, +0.03]
FIBS lin. follow-up	none	−0.124 [−0.18, −0.05]	−0.106 [−0.15, −0.04]	−0.179 [−0.25, −0.10]	−0.182 [−0.24, −0.08]	−0.056 [−0.10, −0.01]
FIBS axial follow-up	full	−0.011 [−0.05, +0.01]	−0.024 [−0.07, −0.00]	−0.032 [−0.24, +0.02]	−0.042 [−0.11, −0.01]	−0.001 [−0.17, +0.04]
FIBS axial follow-up	none	−0.122 [−0.22, −0.00]	−0.119 [−0.19, −0.01]	−0.246 [−0.34, −0.13]	−0.213 [−0.31, −0.04]	−0.134 [−0.21, −0.04]
FIBS linear tokens	full	−0.091 [−0.13, −0.01]	−0.076 [−0.11, −0.01]	−0.133 [−0.18, −0.05]	−0.097 [−0.15, −0.00]	−0.029 [−0.08, +0.01]
Apollo	none	−0.045 [−0.09, +0.02]	−0.051 [−0.08, +0.01]	−0.100 [−0.15, −0.04]	−0.064 [−0.09, +0.02]	−0.047 [−0.09, −0.00]
J-lens mass	none	−0.026 [−0.06, +0.03]	−0.035 [−0.07, +0.00]	−0.054 [−0.10, +0.03]	−0.061 [−0.10, −0.01]	−0.030 [−0.07, +0.01]

Table 5: TPR difference vs the trajectory max at FPR 0.01 (`wb/tables.md`, `wb/mb_paired.csv`). No accumulator beats max for any probe stream (only the leaky supervised probe has non-significant positive cells).

where accumulation won in Idea 1.

3.5 (b) Per-context probe thresholds

	τ at global 5% (p)		context share of honest-max variance	
Stream	full	none	full	none
LLM (`gpt-4.1-mini`)	0.46 (0.16)	0.00 (1.00)	0.002	0.035
`sab`	1.21 (0.004)	0.53 (0.16)	0.075	0.087
FIBS linear follow-up	2.70 (<0.001)	0.95 (0.001)	0.140	0.110
FIBS axial follow-up	1.64 (<0.001)	0.37 (0.27)	0.107	0.068
FIBS linear tokens	0.39 (0.30)	1.26 (<0.001)	0.029	0.048
Apollo deception	3.79 (<0.001)	1.89 (0.001)	0.036	0.103
J-lens mass	0.56 (0.17)	0.31 (0.37)	0.030	0.020

Table 6: Between-cluster heterogeneity of honest false-positive rates on 7 MB task clusters: logit-normal τ at a global 5% threshold, and the share of honest-max variance explained by cluster (`wb/tables.md`, `wb/context_heterogeneity.csv`). Per-cluster honest FPRs range up to 0–24.6% for probes vs 0–12.3% for the LLM.

- **Probe streams are clearly more context-dependent than the LLM.** τ is 1.2–3.8 for four probe streams in full mode (all $p \leq 0.004$) vs 0.46 (n.s.) for the LLM. Cluster share of honest-max variance is 2–14% vs 0.2–3.5%. This replicates the context confound reported for Apollo-style probes, and it is the precondition for per-context allocation to matter.
- **Allocation at 10% spend, spend-matched** (CFAR, global and Stackelberg rows matched in ≥87% of replicates; `wb/context_allocation.csv`). CFAR raises uniform-attacker TPR more for probes than for the LLM, but the only interval excluding 0 is Apollo none, 0.31 → 0.46 [+0.01, +0.29], 1 of 14 MB contrasts (`sab`: full +0.04, none +0.09, n.s.; LLM −0.02 / +0.02). CFAR lowers safety against an informed context-selecting attacker (FIBS linear follow-up full: 0.36 → 0.09, [−0.46, +0.01]). Stackelberg keeps selection safety within 0.02 of global or above on all 14 MB panels (no interval excludes 0). With 7 clusters most differences are n.s. The 5% rows (matched in 40–80% of replicates) and the LinuxArena allocation rows (15–70%, clamped; audit2 m10) are not used.

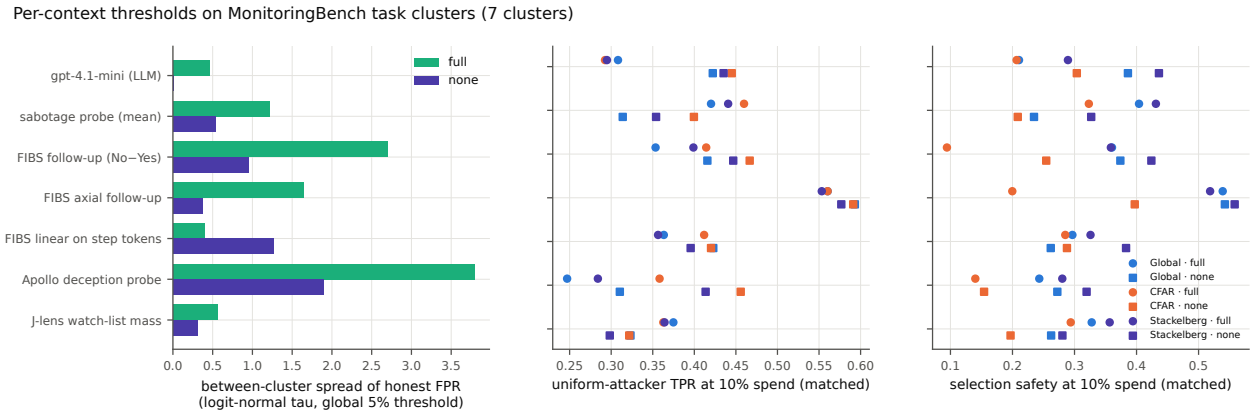


Figure 2: Left: τ per stream and mode. Middle: uniform-attacker TPR at 10% spend (spend-matched). Right: safety against an informed context-selecting attacker, for global, CFAR and Stackelberg thresholds.

- **Related correction** (first audit C1, `results/fixes/corrections.csv`). The earlier E5 claim that "CFAR roughly doubles uniform TPR for probes at 1%" is replaced: 1% spend-matching is not estimable (per-context rules have a realized-spend floor of about 2.5%), and at equal realized spend CFAR gains +0.18 [+0.08, +0.27] for Apollo full (3.6%) and +0.15 [+0.03, +0.28] for FIBS linear follow-up full (2.9%), i.e. 1.3–1.5×, not 2×.

3.6 (c) Fusion and triage with the LLM monitor

Probe	Mode	LLM alone	Probe alone	KDE-LLR (traj)	Rank-sum	Cascade 5%	Cascade 10%
`sab`	full	0.166 (1.4%)	0.258 (1.5%)	0.279 (1.0%)	0.264 (0.8%)	0.306 (1.2%)	**0.299** (1.5%)
`sab`	none	0.151 (1.3%)	0.087 (1.0%)	0.201 (1.7%)	0.199 (1.2%)	0.345 (1.0%)	0.280 (1.0%)
FIBS axial f-u	full	0.166 (1.4%)	0.280 (**2.6%**)	0.328 (**2.8%**)	0.293 (1.9%)	0.336 (1.8%)	0.214 (1.2%)
FIBS axial f-u	none	0.151 (1.3%)	**0.324** (0.9%)	0.312 (1.7%)	0.314 (2.0%)	0.148 (1.3%)	0.150 (1.3%)
FIBS lin. f-u	full	0.166 (1.4%)	0.137 (2.7%)	0.214 (2.8%)	0.207 (1.8%)	0.319 (1.7%)	0.264 (1.4%)
J-lens mass	full	0.166 (1.4%)	0.112 (1.5%)	0.221 (2.5%)	0.224 (1.3%)	0.149 (1.4%)	0.162 (1.8%)

Table 7: TPR at nominal trajectory FPR 0.01, with *realized* FPR in parentheses, plug-in thresholds, 10 splits (`wb/fusion.csv`). "Cascade q": the probe flags the top q of honest-calibrated steps, and only those steps go to the LLM, whose score decides. KDE-LLR: a trajectory-level log-likelihood ratio of (probe max, LLM max). Bold FPRs are the A1 problem: realized FPR about 2× nominal.

- **The `sab` cascade holds.** Full reading, 10% of honest-calibrated steps escalated (13.1% of all steps): TPR 0.288 vs 0.165, +0.123 [+0.059, +0.217], at realized FPR 1.34% vs 1.59% under the conformal rule (plug-in +0.129 [+0.07, +0.23] at 1.43% vs 1.81%), so the cascade does not overspend (`audit2/repro_wb_fusion.csv`, `results/fixes2/wb_a2fusion_conformal_thrfix/`). It is a power gain *and* an 87% cut in LLM calls: the probe removes honest steps the LLM would false-alarm on, so a tighter LLM threshold becomes affordable. Per-step reading gives the same +0.129 [+0.04, +0.18].
- **Cascade caveats.** $q = 10\%$ was one of 5 levels × 7 rules × 5 probes × 2 modes; $q = 5\%$ gives +0.146 and $q = 20\%$ only +0.04 (`audit2/repro_wb_fusion.csv`). The gain is +0.165 on opus-4-5 attacks vs +0.067 on opus-4-6 (`audit2/repro_wb_fusion_strata.csv`).

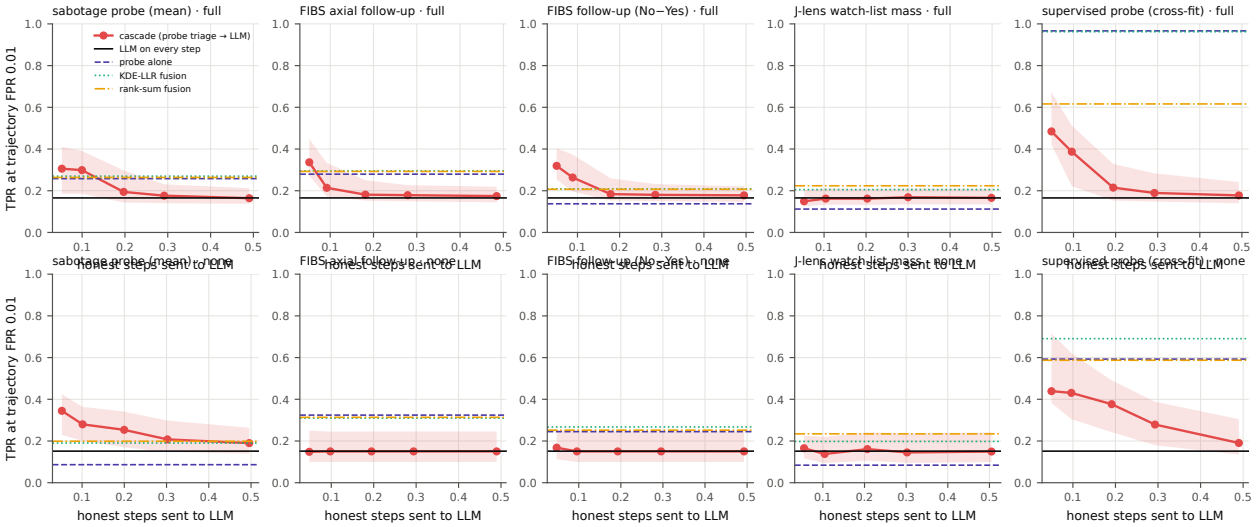


Figure 3: TPR at trajectory FPR 0.01 vs the share of honest steps sent to the LLM (cascade, red), compared with the LLM on every step (black), the probe alone, and KDE-LLR / rank-sum fusion. Top row: full reading. Bottom row: per-step reading.

- **FIBS-axial full fusion headlines are weakened (A1).** KDE-LLR's 0.328 (+0.16) runs at 2.8% realized FPR vs 1.4% for the LLM, so it is not an FPR-matched comparison, and the standalone full-reading FIBS-axial gain vanishes at equal FPR (above). The FIBS-axial-full cascade at 5% has +0.134 [+0.069, +0.276] at realized 1.35% vs 1.59% under the conformal rule (audit-2 code), but q was selected among many cells.
- **Per-step FIBS axial needs no LLM.** The probe alone (0.324 at 0.9% realized FPR) is as good as any fusion. The cascade, in which the LLM decides, falls back to LLM-level TPR.
- **Weak probes still help fusion.** J-lens mass in rank-sum fusion is +0.058 [+0.03, +0.11] at 1.3% realized FPR, although the J-lens alone is below the LLM.

3.7 (d) Probes on detector-aware rewrites

Idea 3's monitor-aware attacker rewrote MonitoringBench attacks against LLM monitors. Round 2 (pacing arms against `gpt-4.1`, plus a matched rewrite control) is fully re-analyzable. Round 1's reader streams were lost (A2) and regenerated in round 4; the pilot's round-1 detection rows reproduce (mean |diff| 0.002).

- **The rewrite artifact is monitor-type specific.** The control rewrite roughly doubles detection by `gpt-4.1` (0.24 → 0.57) and the follow-up probes (FIBS axial 0.38 → 0.65), but leaves token-level probes unchanged (`sab` 0.087 → 0.080; FIBS tokens 0.22 → 0.21). The rewrites condense attacks and carry fabricated tool outputs (first audit M5); monitors that *ask a question about the transcript* react to that, probes on the action tokens do not.
- **Pacing written against `gpt-4.1` transfers partly to the follow-up probe, not to token probes.** One-shot pacing vs one-shot control is −0.05 to −0.09 for `gpt-4.1` and −0.05 to −0.07 (n.s.) for FIBS axial, but +0.02 to +0.09 for `sab` and FIBS tokens. No contrast is Holm-corrected. Only `gpt-4.1` pace_multichart has an interval excluding 0 in the evasion direction.
- **Why the token probes do not see the artifact.** On the E5 panel (first audit, M5) the

	Detection TPR (FPR 0.01, max)			One-shot pace − one-shot control [CI]		
Stream (none)	original	control	pace_max (r0)	pace_max	pace_cusum	pace_multichart
`gpt-4.1` (LLM)	0.240	0.573	0.493	−0.080 [−0.16, +0.03]	−0.053 [−0.11, +0.04]	−0.093 [−0.16, −0.01]
FIBS axial follow-up	0.380	0.653	0.593	−0.060 [−0.13, +0.03]	−0.047 [−0.11, +0.02]	−0.067 [−0.14, +0.02]
FIBS linear follow-up	0.227	0.513	0.500	−0.013 [−0.10, +0.06]	−0.027 [−0.10, +0.06]	−0.007 [−0.09, +0.07]
`sab`	0.087	0.080	0.100	+0.020 [−0.06, +0.09]	+0.020 [−0.03, +0.07]	+0.047 [+0.00, +0.12]
FIBS linear tokens	0.220	0.207	0.293	+0.087 [+0.00, +0.16]	+0.040 [−0.03, +0.11]	+0.053 [−0.02, +0.15]
J-lens mass	0.073	0.300	0.260	−0.040 [−0.13, +0.08]	−0.020 [−0.11, +0.07]	−0.080 [−0.15, +0.02]
Apollo deception	0.113	0.260	0.253	−0.007 [−0.09, +0.07]	+0.027 [−0.04, +0.11]	+0.000 [−0.08, +0.08]

Table 8: T3 round 2 (150 base attacks, per-step reading; `wb/tables.md`, `wb/t3_detection.csv`). "Control" is a rewrite with the same rewriter and no pacing instruction. Pacing arms were written against `gpt-4.1`.

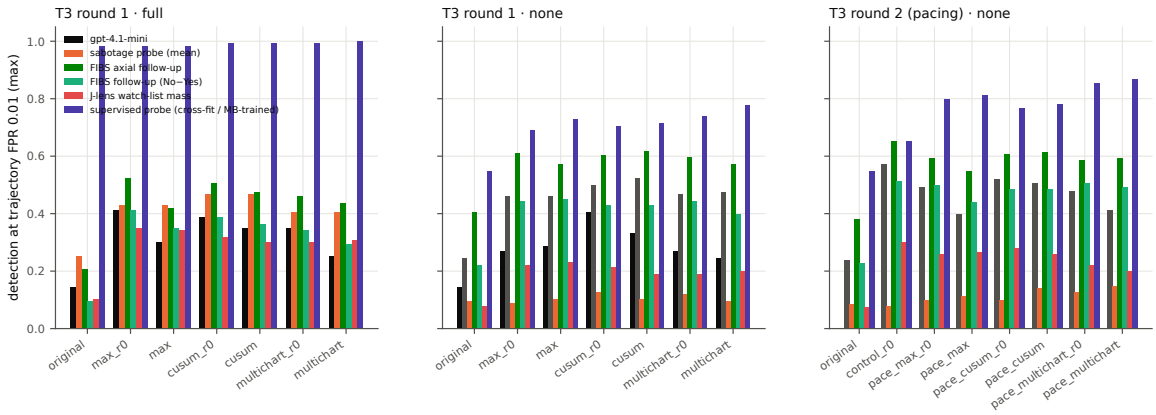


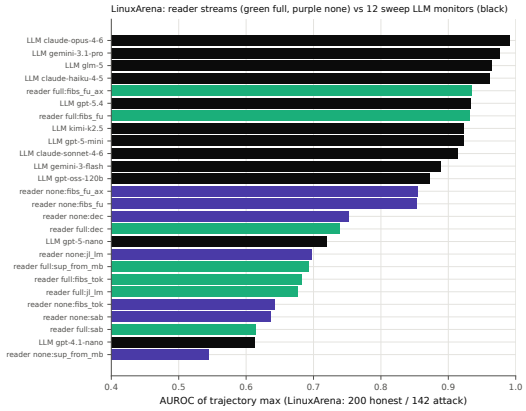
Figure 4: Detection at trajectory FPR 0.01 (max) on T3 rewrites by arm, for the LLM monitors and reader streams. Round 1 (left, middle): pilot streams, lost and regenerated in round 4 with matching results. Round 2 (right) is intact.

same pattern appears as an AUROC confound. Refined variants of the same base attacks score higher than their originals for FIBS linear follow-up full (+0.20 [+0.11, +0.30]) and Apollo full (+0.31). They do not score higher for `sab` max-token (−0.07 [−0.16, +0.01]). On the 33 unrewritten originals the `sab` probes have the highest AUROC (0.81–0.83, vs `gpt-4.1` 0.77) (`fixes/corrections.csv`).

- **Scope.** Every arm is an LLM-monitor-aware *prompted* rewrite. None is a probe-aware or trained evader, so this says nothing about robustness to the attacks in §1.

3.8 LinuxArena

- **FIBS follow-up probes in full reading are comparable to mid-tier frontier monitors.** AUROC is 0.93–0.94, against `gpt-5.4` at 0.933 and `sonnet-4-6` at 0.913. They are below `opus-4-6` and `gemini-3.1-pro`.
- **The pilot ranking had no paired test (audit2 m3).** Its marginal CIs overlap ([0.56, 0.86] vs [0.37, 0.70] for `gpt-5.4`), and the reader's TPR of 0.76 at 1% was at about 2% realized FPR under the plug-in rule ($n_{\text{cal}} = 100$). Round 4 ran the paired comparison (Section 4.2).
- **Mode reversal.** Unlike MB, full reading beats per-step reading here (0.94 vs 0.85).
- **The sabotage probe and J-lens are weak** (AUROC 0.62–0.75). The `sab` contrast pairs



Monitor	AUROC	TPR@0.05	TPR@0.01
LLM `claude-opus-4-6`	0.991	0.943 [0.90, 0.99]	0.923 [0.79, 0.97]
LLM `gemini-3.1-pro`	0.976	0.934 [0.87, 0.98]	0.902 [0.27, 0.95]
Reader FIBS axial f-u (full)	0.936	0.825 [0.75, 0.89]	0.756 [0.56, 0.86]
LLM `gpt-5.4`	0.933	0.702 [0.58, 0.86]	0.568 [0.37, 0.70]
Reader FIBS linear f-u (full)	0.932	0.784 [0.69, 0.88]	0.713 [0.57, 0.82]
LLM `claude-sonnet-4-6`	0.913	0.707 [0.62, 0.83]	0.645 [0.52, 0.75]
Reader FIBS axial f-u (none)	0.854	0.446 [0.13, 0.66]	0.167 [0.04, 0.52]
Reader J-lens hits (full)	0.754	0.419 [0.28, 0.57]	0.297 [0.19, 0.48]
Reader `sab` (full)	0.615	0.262 [0.14, 0.35]	0.159 [0.07, 0.30]
LLM `gpt-4.1-nano`	0.613	0.193 [0.11, 0.29]	0.120 [0.04, 0.22]

Figure 5: LinuxArena (200 honest, 142 attacks), trajectory max. Left: AUROC of reader streams (green full, purple per-step) vs 12 LLM monitors (black) on the same trajectories (`wb/la_auroc.csv`). Right: excerpt with marginal 95% CIs (`wb/tables.md`), pilot, plug-in thresholds. The score files were lost (A2) and regenerated in round 4, which reproduces these AUROCs within 0.001 and adds paired tests (Section 4.2).

were generated from generic objectives, and LinuxArena's side tasks are more diverse.

3.9 Stage 2: policy vs reader activations on MOLE

- **Policy arm.** Qwen3.6-27B, the model that generated MOLE's `qwen36_27b` single-day sessions, reads its own sessions in its chat format, including the system prompt. That prompt states the insider objective. Reasoning tokens and tool schemas are not in the release.
- **Reader arm.** Qwen3.5-27B in the monitor framing, without the system prompt.
- **Probes.** `sab` was re-trained on each model's own activations of the same contrast pairs (CV AUROC 0.97 for both; `EXPERIMENT_LOG.md`). The FIBS 3.5 linear probe is applied to Qwen3.6 as a transfer.

	Policy (Qwen3.6-27B)		Reader (Qwen3.5-27B)	
Stream	max over turns	mean over turns	max over turns	mean over turns
`sab` (layer mean)	0.839 [0.80, 0.88]	0.954 [0.94, 0.97]	0.863 [0.83, 0.89]	0.955 [0.94, 0.97]
`sab` L43 (post hoc)	0.899 [0.87, 0.93]	0.978 [0.96, 0.99]	0.938 [0.92, 0.96]	0.982 [0.97, 0.99]
FIBS linear on tokens	0.816 [0.78, 0.86]	**0.982** [0.97, 0.99]	0.620 [0.57, 0.67]	0.742 [0.70, 0.79]
J-lens mass	0.683 [0.64, 0.73]	0.764 [0.72, 0.80]	0.752 [0.71, 0.80]	0.829 [0.79, 0.87]
J-lens top-25 hits	0.800 [0.76, 0.84]	0.925 [0.90, 0.95]	0.956 [0.94, 0.97]	**0.984** [0.97, 0.99]

Table 9: Session AUROC, 233 insider vs 200 random benign sessions (`wb/mole_policy_vs_reader.csv`). TPR at 1% FPR for `sab` mean: policy 0.71, reader 0.74.

- **Policy activations do not beat the reader on the main probe** (`sab`: 0.84 / 0.95 vs 0.86 / 0.96).
- **The one stream where the policy wins is suspect.** FIBS on tokens gives 0.98 vs 0.74 (mean), but this is also the stream that could key on the in-context insider instruction, which only the policy sees.

- **The reader's J-lens hit count is the best stream of either arm** (0.96 / 0.98).
- **The pilot split cannot answer the policy-vs-reader question cleanly.** There is no LLM baseline for it; benign sessions are a random sample whose account and role mix is not matched to the insiders; and the policy sees a system prompt that names the objective. Round 4 fixed all three (Section 4.3); with matched negatives the reader detects more than the policy at 5% FPR.

4 Round 4: preregistered replication, regenerated streams, matched MOLE

Track `r4_wb` (October 22–23). The preregistration and the fixed held-out sets were committed before any held-out item was scored; the first held-out score was written 17 s after that commit (audit 3). Reader code, model revision and probes are unchanged from the pilot; honest trajectories re-read on the new pods agree with the pilot's stored streams at bf16-noise level (Honest-550 `none`: FIBS axial $r(\max) = 0.9993$; every mean-pooled stream $r \geq 0.999$) (`results/r4_wb/agreement_*.csv`).

4.1 Part 1: replication of the MB headline [C]

Design. H1: TPR(FIBS axial follow-up, `none`) − TPR(`gpt-4.1-mini`, `none`) > 0 at trajectory FPR 1% on held-out attacks; supported if the 95% CI lower bound is > 0. Attacks: 1,000 MB attacks that the pilot never used (771 opus-4-5, 229 opus-4-6), stratified by agent model, structure and side task. All S4 (distributed) attacks were in the pilot, so the replication covers S1–S3 only. Honest: Honest-550, re-used as preregistered (the honest side is not new data; 973 of the 1,000 attacks share a task with it); reader honest scores recomputed on the replication pod. 200 task-grouped splits; paired task-cluster bootstrap with calibration noise ($B = 2{,}000$). Holm family S1–S3.

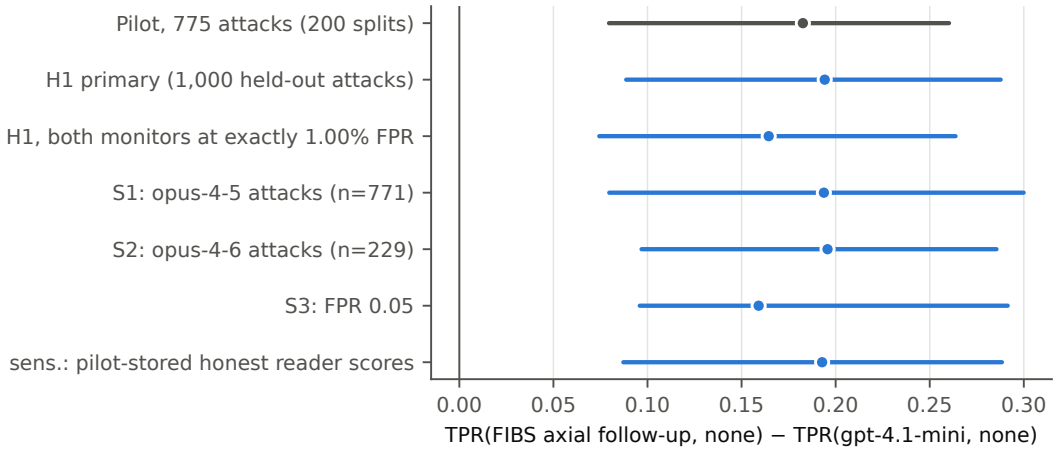


Figure 6: Round-4 replication, conformal thresholds: reader minus LLM TPR with 95% CIs. Grey: the pilot estimate on its 775 attacks (200 splits). Sources: `results/fixes2/wb_r4primary_conformal_thrfix/primary.csv`, `results/fixes2/wb_a3primary_conformal_thrfix/wb_primary.csv`, `results/fixes2/wb_r4a1_conformal_thrfix/a1_mb_equalfpr.csv`; drawn by `docs/reports/src/make_r4_report_figs.py`.

- **H1 is confirmed.** The lower bound is $+0.089 > 0$, and the estimate lies inside the pilot

Contrast (FPR 0.01 unless stated)	TPR reader	TPR LLM	realized FPR reader / LLM	Δ [95% CI]	Holm
H1 primary	0.258	0.064	1.24% / 1.14%	**+0.194** [+0.089, +0.288]	–
H1, both at exactly 1.00% realized FPR (audit 3)	0.220	0.056	1.00% / 1.00%	+0.164 [+0.075, +0.263]	–
S1: opus-4-5 attacks ($n = 771$)	0.250	0.056		+0.194 [+0.080, +0.300]	< 0.001
S2: opus-4-6 attacks ($n = 229$; agent not in Honest-550)	0.287	0.091		+0.196 [+0.097, +0.285]	< 0.001
S3: FPR 0.05	0.419	0.260	5.85% / 5.23%	+0.159 [+0.096, +0.291]	< 0.001
sens.: pilot-stored honest reader scores	0.257	0.064		+0.193 [+0.087, +0.288]	
sens.: 972 attacks with complete LLM streams	0.246	0.059		+0.187 [+0.085, +0.283]	
sens.: missing LLM steps re-scored	0.258	0.066		+0.192 [+0.087, +0.286]	
exploratory: `sab` (none) − LLM	0.038	0.064		−0.026 [−0.099, +0.018]	
exploratory: FIBS linear follow-up (none) − LLM	0.138	0.064	1.60% / 1.14%	+0.074 [+0.002, +0.145]	

Table 10: Round-4 replication under the conformal rule (`results/fixes2/wb_r4primary_conformal_thrfix/primary.csv`; equal-FPR row from `results/fixes2/wb_a3primary_conformal_thrfix/`). Plug-in: H1 +0.203 [+0.100, +0.296] at realized 1.39% / 1.28%; both-at-1.00% +0.167 [+0.065, +0.256].

interval. The preregistered equal-FPR guard was not triggered (realized-FPR gap 0.10 pp < 0.25 pp). With both monitors bisected to exactly 1.00% realized FPR the gain is +0.164 [+0.075, +0.263]. An independent re-implementation gives +0.192 [+0.095, +0.283] (audit 3, conformal).

- S1–S3 survive Holm. AUROC of the trajectory max: 0.817 reader vs 0.715 LLM.
- Both TPRs are lower than in the pilot (0.32 / 0.14), because the pilot was S4-heavy; the gap between the monitors is not.
- **Strata** [E] (descriptive, unadjusted): every structure and side task is positive (S1 +0.17, S2 +0.13, S3 +0.21; malware +0.32, sudo +0.27, passwords +0.18, firewall +0.07). Firewall attacks are nearly undetectable by both monitors (0.08 / 0.01).
- **External honest runs** [E] (200 BashArena opus-4-6 honest runs on tasks not in Honest-550): at the Honest-550 thresholds the realized FPR at 1% is 1.06% for the reader and 1.17% for the LLM, i.e. about 2 alarms in 200 for each. That shows no worse transfer for the reader but cannot rank the two (audit-3 gap interval $[-1.5, +1.1]$ pp). The round-4 log's "transfers at least as well" is withdrawn.
- **Scope.** The comparator is one cheap LLM monitor (`gpt-4.1-mini`, AUROC 0.72 here), not a frontier monitor; the attacks share scenarios with the pilot (762 of 1,000 share task × side task × agent with a pilot attack); and the FIBS probe's training data may include BashArena-like transcripts (FIBS is an external release, so disjointness cannot be verified). Of the other generic probes, `sab` is null and the FIBS linear follow-up is borderline.

4.2 Part 2: LinuxArena paired comparison and equal-FPR checks

Reproduction. LinuxArena (342 trajectories) and T3 round 1 (509) were re-read in both modes. The pilot's tables reproduce: LinuxArena AUROC within 0.001 for every stream, TPR at 0.01/0.05 within 0.008, T3-r1 detection rows mean |diff| 0.002 (`results/r4_wb/repro/`).

Paired comparison [C]. FIBS axial follow-up (`full`) minus each of the 12 sweep monitors on the trajectories both scored (200 honest, 101 attacks); 200 splits grouped by main task; paired bootstrap; Holm over the 12 contrasts at each α. With $n_{\text{cal}} = 100$ honest runs, a 1% threshold rests on about one calibration point.

- **Nominal 1%, conformal rule.** The reader's realized FPR is 0.84–1.09% and its TPR 0.66–0.72 (under the plug-in rule it ran at 1.8–2.1% with TPR 0.76–0.81, the pilot's "0.756"). It is Holm-significantly above 7 of 12 monitors with the committed split code (gpt-4.1-nano,

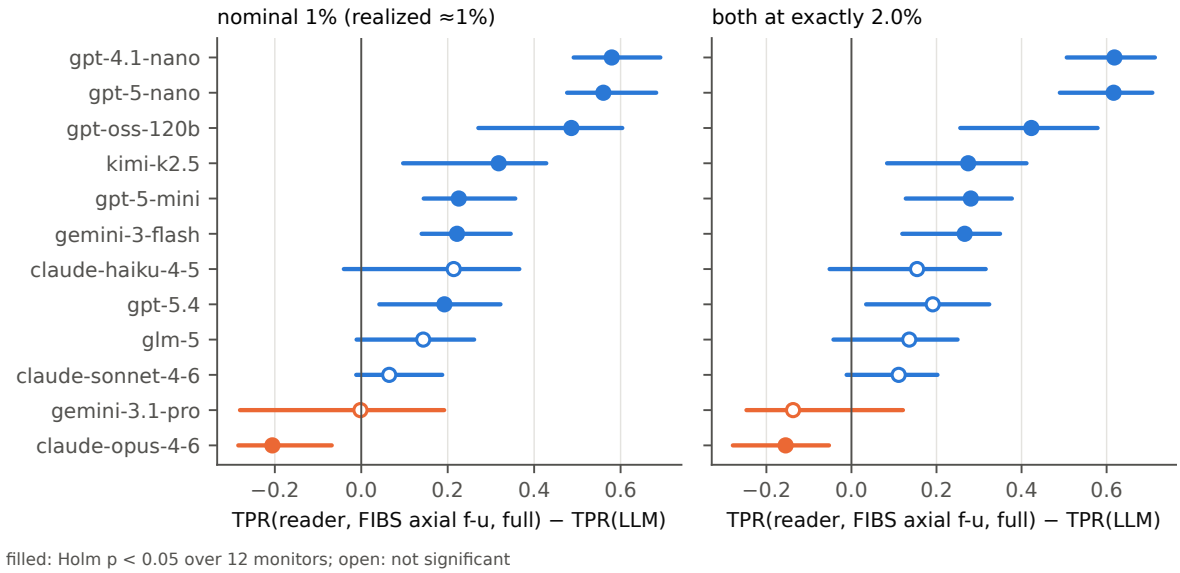


Figure 7: LinuxArena: reader (FIBS axial follow-up, full) minus each LLM monitor, TPR of the trajectory max, conformal thresholds. Left: nominal 1% (committed analysis code; realized FPR reader 0.84–1.09%, LLMs 0.90–1.20%). Right: both monitors bisected to exactly 2.0% realized FPR (audit-3 code). Filled markers: Holm $p < 0.05$ over 12 monitors. Sources: `results/fixes2/wb_r4la_conformal_thrfix/la_paired.csv`, `results/fixes2/wb_a3la_conformal_thrfix/wb_la_equalfpr.csv`.

gpt-5-nano, gpt-oss-120b, kimi-k2.5, gpt-5-mini, gemini-3-flash and gpt-5.4, the last at Holm $p = 0.035$) and 6 of 12 with the audit's split code (gpt-5.4 at Holm $p = 0.055$). It is not distinguishable from haiku-4.5, glm-5, sonnet-4.6 and gemini-3.1-pro, and is significantly below opus-4.6 (-0.21 [-0.29, -0.07]).

- **Equal realized FPR.** The preregistered equal-FPR comparison was not run in round 4 (an undeclared omission); audit 3 ran it. At exactly 1.00% for both monitors the count is rule-dependent: 7 of 12 above and 1 below under the plug-in rule, 10 of 12 above and 0 below under the conformal rule, because 8 of the 12 LLM levels fall below $1/(n_{\text{cal}} + 1)$, where the conformal rule never alarms with certainty. This comparison sits at the calibration resolution limit and is not reported as a count. At exactly 2.0% the result is the same under both rules: 6 of 12 above (gpt-4.1-nano, gpt-5-nano, gpt-5-mini, gpt-oss-120b, kimi-k2.5, gemini-3-flash) and 1 below (opus-4.6), with reader TPR 0.76–0.81.
- At 5% (nominal, conformal) the reader is above 5 monitors (sonnet-4.6 at Holm 0.036) and below gemini-3.1-pro (-0.11) and opus-4.6 (-0.12).
- In `none` mode the reader is weak on LinuxArena (TPR ≤ 0.18 at 1%) and loses to most monitors [E].
- **Revised claim.** In full reading the reader probe is above the small and mid-size monitors, not distinguishable from haiku-4.5, glm-5, sonnet-4.6 and gemini-3.1-pro, and below opus-4.6.

Equal-FPR check on the pilot MB data (A1) [E]. Full-reading FIBS axial: the earlier "+0.061 [+0.021, +0.219] at equal realized FPR" was the gain at the plug-in FPR floor (1.6–1.7%) with a bootstrap bias of about +0.04. Under the conformal rule the reader can be lowered to the LLM's FPR, and the gain is -0.015 [-0.055, $+0.097$]; with both at exactly 1.00%, -0.019 [-0.057, $+0.087$]. **This claim is dropped.** FIBS axial `none` (+0.183 [+0.080, +0.260]) and `sab` full (+0.093 [+0.012, +0.161]) hold at equal FPR (`results/fixes2/wb_r4a1_conformal_thrfix/a1_mb_equalfpr.csv`).

Per-context calibration is not a usable fix at 1% [E]. Per-cluster plug-in thresholds on the 8 MB task clusters realize 5–7.6% FPR at nominal 1% for every monitor, and about 10% on the 10 LinuxArena environments (10 honest runs per environment per calibration half). A global threshold reported with its realized FPR remains the right comparison.

4.3 Part 3: MOLE policy arm with account-matched negatives [C]

Design. 233 insider sessions of the `qwen36_27b` single-day corpus, each paired with a benign session of the *same account on the same day* (38 accounts). Arms: `policy` (Qwen3.6-27B reads its own sessions with the original system prompt, which states the insider objective); `policy_neutral` (the attack session's system prompt replaced by its paired benign session's, so the instruction is identical within a pair); `reader` (Qwen3.5-27B, monitor framing); `gpt-4.1-mini` on the reader-format transcript, one call per session (insider-aware and generic prompts). Probe statistic: `sab`, max over turns. Preregistered Holm family of five contrasts at session AUROC, and the same five at TPR at 5% session FPR; account-cluster bootstrap ($B = 2{,}000$), paired across arms.

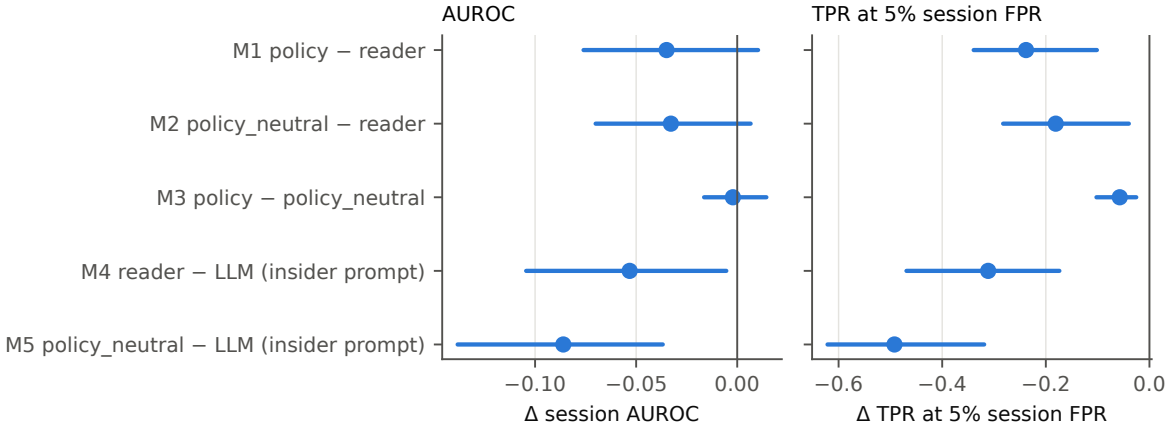


Figure 8: MOLE, account-matched negatives: the five preregistered contrasts as differences in session AUROC (left) and in TPR at 5% session FPR (right, conformal thresholds), account-cluster 95% CIs (`results/fixes2/wb_a3mole_conformal_thrfix/wb_mole.csv`). The TPR family was preregistered but omitted from the round-4 log; audit 3 computed it.

Contrast	Δ AUROC [95% CI] (Holm p)	Δ TPR at 5% FPR [95% CI] (Holm p)
M1 policy − reader	−0.035 [−0.076, +0.010] (0.30)	**−0.238** [−0.339, −0.101] (< 0.001)
M2 policy_neutral − reader	−0.033 [−0.070, +0.007] (0.30)	−0.181 [−0.282, −0.040] (0.008)
M3 policy − policy_neutral	−0.002 [−0.016, +0.014] (0.84)	**−0.057** [−0.102, −0.025] (0.008)
M4 reader − LLM (insider prompt)	−0.053 [−0.104, −0.005] (0.14)	−0.311 [−0.469, −0.174] (< 0.001)
M5 policy_neutral − LLM (insider prompt)	−0.086 [−0.138, −0.037] (0.015)	−0.492 [−0.621, −0.319] (< 0.001)

Table 11: MOLE contrasts (audit-3 implementation, conformal thresholds for TPR). TPR at 5% FPR: reader 0.47, policy 0.23, policy_neutral 0.28, LLM (insider prompt) 0.78, at realized FPR 5.1–5.4% (`results/fixes2/wb_mole_fpr_thrfix.csv`). Session AUROCs: policy 0.815, policy_neutral 0.817, reader 0.850, LLM 0.903 (insider prompt) / 0.915 (generic) (`results/r4_wb/mole_arms.csv`).

- **AUROC shows no policy-vs-reader difference** (M1–M3 n.s.). **At 5% session FPR the reader beats the policy**: 0.47 vs 0.23, −0.24 [−0.34, −0.10]. The round-4 log's "policy

activations do not beat the reader" is therefore corrected: they are worse at the preregistered operating point.

- **The insider instruction lowers the policy probe's TPR**: −0.057 [−0.102, −0.025] (Holm 0.008). The round-4 log's "removing the instruction changes nothing measurable" holds only for `sab`-max AUROC (shift bounded to about ±0.017). On other statistics the instruction matters too [E]: mean over turns +0.014 [+0.006, +0.024], FIBS-on-tokens max −0.085, J-lens mass max −0.060. "The probes key on content, not the instruction" over-generalizes from one statistic.
- **A cheap LLM session monitor beats every `sab`-max probe arm** (M5 Holm 0.015; M4 nominal only at AUROC). The LLM reads the whole session while the probes take a max over turns, so this compares aggregation as much as signal. With the mean-over-turns statistic the reader is above the LLM in AUROC (+0.050 [+0.008, +0.098], secondary [E]).
- **"Unmatched negatives explain the pilot AUROCs" fails.** With matched negatives the mean-statistic AUROCs are still 0.94–0.96 (pilot 0.95–0.96). The pilot's high AUROCs come from the mean statistic, which was chosen post hoc.
- Reader J-lens top-25 hits: 0.935 (max) / 0.974 (mean), the best probe stream [E].

4.4 Reasoning leakage (audit 3)

Agent reasoning is nearly label-revealing in these datasets (on MB, honest runs show reasoning on every step and attacks on 43%), so a monitor that saw it could separate the classes for the wrong reason. No monitor scored in this repository receives it: the white-box reader export, the `gpt-4.1(-mini)` monitors in every history mode and the MOLE renders all exclude agent reasoning, and no text-only assistant turn appears among the visible MB steps (`results/audit3/leak_reasoning.csv`). The MOLE session arms show assistant turn content by design, equally in every arm. The LinuxArena sweep scores were computed externally and cannot be checked.

4.5 Deviations and caveats

- OpenRouter credit ran out mid-run: 305 LLM steps in 28 held-out attacks were missing in the registered run and handled by `nanmax` as preregistered; re-scoring them later changes Δ by −0.002. The 200 external honest runs were scored after credit returned.
- `analyze_primary.py` was finalized after the data arrived but matches the preregistration (audit 3). The preregistration's "1,869 attacks no earlier track scored" is not exact (another track had trajectory-level scores for some; not used for selection).
- The Qwen3.6-27B revision is unpinned, as in the pilot.
- GPU: four on-demand pods, about $70; OpenRouter $34.47.

5 What holds, what is weakened, what is dropped

Table 12 lists every headline claim with its verdict after the three audits (`results/fixes/`, `results/audit2/`, `results/audit3/`) and the threshold-convention re-run (`results/fixes2/`).

Errata relative to the October 22 version

- Summary item 1: pilot gain +0.19 to +0.21 (0.34 vs 0.15) is now +0.18 to +0.21 (0.32 vs 0.14) under the conformal rule, and is confirmed on held-out attacks.

Claim	Verdict	Basis
Per-step FIBS axial follow-up beats `gpt-4.1-mini` at 1% FPR	**confirmed** [C]	Held-out replication +0.194 [+0.089, +0.288]; +0.164 at exactly 1.00% FPR for both; S1–S3 only; one cheap comparator
`sab` cascade (10%) beats the LLM with 87% fewer calls	**holds** [E]	+0.123 [+0.059, +0.217] at realized FPR 1.34% vs 1.59%; q chosen among 5; larger on opus-4-5
No accumulation rule beats max on probe streams	**holds** [E]	Table 5; consistent with Idea 1
Probe streams are more context-heterogeneous than LLM streams	**holds** [E]	honest-only statistic
LinuxArena: full-reading FIBS follow-up above small/mid monitors	**holds, revised** [C]	6–7 of 12 Holm-significant at nominal 1% (realized ≈1%, TPR 0.66–0.72); 6 of 12 at equal 2%; below opus-4.6; equal-1% count rule-dependent
Full-reading `sab` beats the LLM (+0.10)	weakened [E]	Holm $p = 0.22$; +0.148 on opus-4-5 vs +0.016 on opus-4-6
Full-reading FIBS axial beats the LLM at equal FPR (+0.061)	**dropped**	−0.015 [−0.055, +0.097] under the conformal rule; the earlier value sat at a plug-in FPR floor
FIBS-axial-full fusion (KDE-LLR 0.328, +0.16)	weakened	realized FPR 2.8% vs 1.4%
Reader threshold transfers to new tasks "at least as well" as the LLM's	**withdrawn**	about 2 alarms in 200 for each; cannot rank
MOLE: policy activations do not beat the reader	**corrected** [C]	No AUROC difference, but at 5% FPR the reader detects more (−0.24)
MOLE: the insider instruction has no measurable effect on the policy probe	**corrected** [C]	Holds for `sab`-max AUROC only; TPR at 5% −0.057 (Holm 0.008)
MOLE: pilot AUROCs came from the mean statistic and unmatched negatives	**half fails**	Matched mean-statistic AUROCs still 0.94–0.96
MOLE: an LLM session monitor beats the max-over-turns probe arms	holds [C], confounded	M5 Holm 0.015; whole-session vs max-over-turns
CFAR on probes "roughly doubles" uniform TPR (E5, 1%)	dropped / replaced	1% matching not estimable; 1.3–1.5× at equal realized spend
`sab` L43 (+0.24)	exploratory only	selected on the evaluation set; not replicated
DiD of CUSUM gain (`sab` +0.16; FIBS −0.18)	exploratory	10 splits, not in a declared family
No monitor sees agent reasoning	holds	audit 3, all monitors scored in this repository

Table 12: Claim ledger after three audits and the threshold-convention re-run.

- Summary item 2: cascade +0.13 is +0.123 [+0.059, +0.217] (conformal).
- Full-reading FIBS axial: "not shown" becomes dropped at equal FPR (−0.015).
- LinuxArena: "no paired test, score files lost" is superseded by the regenerated streams and paired tests (Section 4.2).
- MOLE: "policy activations do not beat the trusted reader" is corrected by the matched design (Section 4.3).
- Thresholds: plug-in values are labelled; headline numbers use the conformal rule.

6 Interpretation

1. **A trusted reader buys some power, not only cost savings.** The literature note expected "white-box adds cost savings, not power, for a trusted reader". One off-the-shelf probe (FIBS axial) on a per-step follow-up question beats a cheap LLM monitor by about 0.16–0.19 TPR at 1% FPR with no task-specific training, and this replicated on new attacks under a preregistration. The comparator is weak (`gpt-4.1-mini`, AUROC 0.66–0.76), so "beats a cheap LLM monitor" is the right scope; on LinuxArena the reader is above small and mid-size monitors and below the strongest (opus-4.6).
2. **The probe's best use with an LLM is triage.** The `sab` cascade gains about 0.13 TPR *and* removes about 87% of LLM calls, the shape of the published GDM and Anthropic misuse cascades[†], here at matched trajectory FPR for multi-step sabotage.
3. **Probe streams need careful calibration.** Their between-task heterogeneity is much larger (τ 1.2–3.8 for four probe streams vs 0.46 for the LLM), and the full-reading FIBS axial threshold did not transfer across tasks (about 2% realized FPR at nominal 1% under either rule). Per-context thresholds are not a fix at 1% with these honest-set sizes. Compare monitors at equal realized FPR with a threshold rule whose FPR is exact.
4. **The decision layer transfers unchanged.** Max is the right trajectory rule on every probe stream.
5. **Token-level probes and question-based monitors fail differently.** Rewriting moves LLM monitors and follow-up probes but not token probes, an argument for combining the two kinds. It is not robustness evidence: no attacker here knew about the probes.
6. **Reading one's own activations is not better than a trusted reader on MOLE.** With matched negatives the policy's probe is no better in AUROC and worse at 5% FPR, and its TPR depends on the instruction in its context.

7 Limitations

- **One reader, one MB comparator** (`gpt-4.1-mini`); no frontier LLM monitor was run on MB. The replication covers S1–S3 only and re-uses Honest-550.
- **Statistical power.** 550 honest runs put a 1% trajectory FPR on about 5 false alarms per calibration half; the project analysis used 10 splits; 7 MB clusters are too few for per-context significance.
- **Off-policy reading.** The reader reads Opus-authored text, the regime where intent probes generalize worst, and shares an LLM monitor's threat model (it sees only text).
- **Label-correlated agent model.** Honest runs are all opus-4-5; like-for-like results are reported, but surface features (truncation, reasoning visibility) remain label-correlated (first audit m8).
- **Calibration resolution.** With 100 LinuxArena honest calibration runs, a 1% threshold rests on about one run, and exact equal-FPR comparisons at 1% are rule-dependent.
- **No adaptive white-box attacker** (every robustness concern in §1 stands untested), and **bf16 noise** in max-pooled features (up to 2 SD for single steps).

8 Recommended next steps

1. **Add a strong LLM comparator on MB** (`gpt-4.1` or a frontier monitor on the same trajectories) to test whether the confirmed FIBS-axial gain survives against a non-trivial

monitor; add S4 attacks from a fresh source.

2. **Replicate the `sab` cascade and `sab` L43** on a fresh split with a pre-declared q and layer.
3. **Fresh open-policy rollouts** (e.g. Qwen3.6-27B on BashArena) with reasoning retained, so policy and reader arms see identical text and the LLM baseline can be history-matched.
4. **Run a probe-aware attacker**: first prompted ("a white-box probe on a trusted model reads your actions") against a strong LLM monitor; then, if stage 1 holds, a trained RL-Obfuscation-style evader.
5. **Larger honest sets** for LinuxArena-type comparisons, so that 1% thresholds rest on more than one calibration run.
6. **Contact FAR.AI** for FIBS's on-policy BashArena and SHADE step streams; our aggregation, per-context and cascade analyses would run on them unchanged.

Cost and reproduction

Pilot: GPU ≈$70 (two on-demand H200 pods, about 7.0 h and 5.8 h at $5.29/h, plus $2.1 of earlier setup; about 1,000 tokens/s per reader process with per-step follow-ups); OpenRouter $2.95 (contrast pairs). Round 4: GPU ≈$70 (three H200 pods and one H100 pod, deleted after outputs were copied off and verified); OpenRouter $34.47. Audits and the threshold re-run $0. Code: `tracks/wb_whitebox/` (reader and analyses), `tracks/r4_wb/` (round 4, incl. `PREREG.md`), `tracks/audit2/`, `tracks/audit3/` (`wb_*`, `leak_*`), `tracks/fixes2/` (`wb_*`); round-4 report figures `docs/reports/src/make_r4_report_figs.py`. Results: `results/wb/`, `results/r4_wb/`, `results/audit2/`, `results/audit3/`, `results/fixes2/`, errata `results/fixes/corrections.csv`. Score streams: `data/processed/whitebox/scores/` (LinuxArena and T3 r1 regenerated in `scores/r4/`). Log: `EXPERIMENT_LOG.md` (2026-10-21/22/23). Repository: https://github.com/gregkocher/agent-monitor-detection (private).